



Technische Hochschule
Ingolstadt

Bachelor Thesis

Faculty of computer science thesis.tex

**Development of a Frontend Application for Real-Time and
24-Hour Monitoring and Recording of Electrocardiographic Data
with Communication via Bluetooth Low Energy**

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AFFIDAVIT

I hereby declare that I have written this thesis independently, have not used any sources or aids other than those stated, have not submitted it elsewhere for examination purposes and have expressly identified all quotations from the sources used, either verbatim or in spirit. I have marked verbatim and indirect quotations as such.

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This is the sentimental part where you get to thank all the persons who were a part of your thesis journey in one or the other way!

Eric Horacsek
Ingolstadt, Deutschland
November xx 2024

ABSTRACT

The human heart is the central organ of the circulatory system and plays a vital role in supplying the body with oxygen and nutrients. Rhythmic contractions pump blood through the body, laying the foundation for all vital functions. An essential part of medical diagnostics combined with heart function and health is the electrocardiogram (ECG). The ECG measures the electrical activity of the heart and provides valuable information about the heart rhythm as well as possible irregularities or diseases. As technology has advanced, the way in which ECG data can be collected and monitored has also evolved significantly. Mobile health monitoring in particular now makes it possible to record ECG data not only at certain points of time, but also in real time and over longer periods of time. Wireless transmission via Bluetooth Low Energy (BLE) enables reliable and energy-efficient transfer of data to mobile devices. This opens up new possibilities for applications offering real-time and long term monitoring enabling users access to relevant information and graphical processing and alarm functions. This thesis explains the conception and implementation of an Android frontend application, which realises continuous and mobile ECG monitoring. The main programming language used for this project, is kotlin with Jetpack Compose as the Framework. Moreover this dissertation is particularly focused on implementing the communication and data transmission via BLE as well as realising an 24-hour-mode, which might be used for 24-hour data recognition. It also provides an broad insight into the given requirements, designprincipals, and technical challenges which accured during the implementation. Furthermore this theseis offers a glimpse into the potential for modern healthsuveillance.

CONFIDENTIALITYCLAUSE

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(Signature)
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GLOSSARY

Eric Horacsek Absolute operation.

ACRONYMS

D Dimensions.

RGB Red Green Blue channels.

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1 INTRODUCTION

This chapter will give a small introduction on what the thesis and the implemented application is about. This covers an explanation for the application's motivation and the environment the application will be used in.

1.1 Motivation

The aim of this work is to create a new version of an Android App once developed by fellow students from the faculty of Electrical and Information Technology. It's core functionality is to receive and display ECG data in real time, via BLE. With the completion of this work, the new version will be able to receive and display ECG data in real time, via BLE. In addition it will be capable of an 24-hour mode which records ECG data for up to 24 hours, so the collected data can be analyzed later.

1.2 Scope and Framework

This paper only deals with the app development process of the new version. Therefore, any implementation of the previous work done by the students is not going to be discussed in detail. The hardware implementation is also not an element of this work . Generally speaking, this paper covers all topics that are essential for the development of the new app version. Specifically, technology stack of choice, IDE choice desicion, communication with hardware via Bluetooth low energy, structure of the user interface, real-time data monitoring, implementing the 24 hour mode. In addition, the outlook explains procedures for the implementation of future features that could not be realized within the limits of this paper.

1.3 Procedure

2 TECHNICAL FUNDAMENTALS

"The nature of the application and the required functionality of a software system should influence the choice of methods and tools used during development". [Clarke \[1995\]](#) This ensures that the development process is both efficient and tailored to meet the unique demands of the system. Moreover, aligning tools and methods with the application's requirements can help streamline workflows and reduce potential complexities during implementation. Consequently choosing the right programming language, aswell as IDE is essential to enable efficient development.

The following chapter Technical Fundamentals contains all technical characteristics and requirements, and how they have been fulfilled.

2.1 Selected Technologies

2.1.1 Programming Language and Framework

When searching for a suitable programming language to program Android apps, names like Java/Kotlin and Dart often appear. Both languages offer great frameworks, such as Flutter for Dart or Jetpack Compose for Kotlin. One aswell as the other are highly effective because they eliminate the limitations of XML-based UI development, enabling a more streamlined, declarative approach that combines UI and logic in a single language. Thus reduces boilerplate code, and enhances performance aswell as developer productivity. The main difference between both frameworks lies in resource management. For example Flutter is superior in cpu usage, but Jetpack Compose consumes less memory. [Kusuma et al. \[2023\]](#) "But since both have their respective advantages and disadvantages, it is impossible to say that one is superior to the other. The choice between Jetpack Compose and Flutter depends on the needs and preferences of the developer. If developers are more familiar with the Kotlin language and want to use new technologies, then Jetpack Compose might be a better choice." [Kusuma et al. \[2023\]](#) With prior practical experience in Java/Kotlin, Jetpack Compose was chosen as the framework for this work.

2.1.2 Development Environment

Integrated development Environments play a fundamental role in today's application development, providing developer with tools mandatory to debug, refactor and write code. Making programming more efficient and delightful for programmers. When developing Android Apps with state of the art frameworks like Jetpack Compose, Android Studio stands out as the most suitable option as both Jetpack compose and Android Studio are

created by Google. Android Studio is the official IDE for Android Development. With IntelliJ as base it inherits powerful code editor and developer tools, combined with Android specific features. Such as Live Edit (updates running app when code is changed) as well as Interactive Preview which allows to test functionality between so called Compose components (Compose will be explained in the later course of this work)¹, et cetera.

The version of Android Studio used for this thesis is **Ladybug | 2024.2.1**. Which is the latest version as this was written (13.12.2024).² Downloading the right version can be crucial for problem-free debugging. As some versions lead to bugs, causing the screen to freeze.³

2.1.3 Versions

This section presents fundamental versions of the required development resources used for this work.

The tests for this application during the development process, were done on a phone using the **Android version 13** which implies the usage of **API level 33**. Utilizing lower Android version than 12 (API Level 31,32) is not possible due to new runtime permissions added in Android 13, making it impossible to start the app⁴. Higher Android versions were not reviewed. More versions of the used resources, are presented in table 1.

Table 1: Versions of used resources

Ressource	Version used
Android version	13
API Level	33
Koltin compiler extension version	5.15.1
minSDK	30

2.2 Android architecture

To program a platform-specific application, it can be important to understand the basic architecture of the platform, especially when dealing with tasks regarding memory or Bluetooth. This section will give a brief insight into the structure of the Android construction. Tasks discussed later in this thesis may reference this chapter to convey a better understanding.

¹<https://developer.android.com/studio/intro>

²<https://developer.android.com/studio/releases>

³<https://stackoverflow.com/questions/78125635/processing-classes-for-emulated-method-breakpoints-this-popup-comes-and-loadin>

⁴<https://developer.android.com/develop/connectivity/bluetooth/bt-permissions>

The core of the Android Platform is a Linux based kernel. On top of Android's Kernel are four more stack layers, such as the Hardware Abstraction Layer, Native Libraries/Android Runtime, Java API Framework and the uppermost layer, the System App layer. (also see: Figure 1). Going into detail for each stack layer is not necessary for this work, consequently only essential layers like the Java API Framework will be considered in more depth.

Table 2: Layers explanation

Layer	Functionality
System App's	This layer is a collection of preinstalled applications. Apps contained in this collection, have the same status at system level as third-party apps(system settings is the only exception). Giving the user freedom for design.
Java API Framework	33
Native Libraries	5.15.1
Android Runtime	30

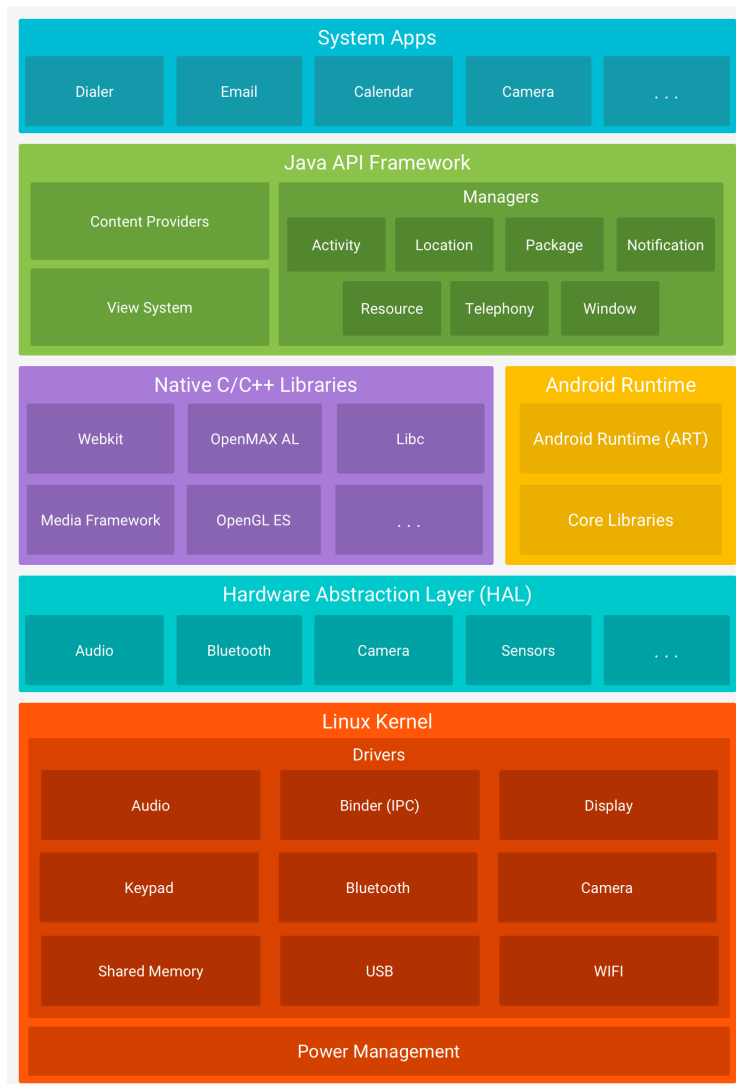


Figure 1: Android Platform architecture

3 DISKUSSION

Diskussion der Resultate im Vergleich zu Ergebnissen der Related work-Analyse und Reflektion auf die Forschungsfragen/Hypothesen.

Hypothese **H1** kann angenommen werden. Bla bla Begründung/Argumentation.

Andererseits zeigen die Ergebnisse der Evaluation aber keinen signifikanten Unterschied (siehe **Field**) zwischen X und Y - somit kann auch **H2** angenommen werden. Da hier bla bla Argumentation.

Durch die Ergebnisse des Fragebogens wird Hypothese **H3** nicht bestätigt. Es sind keine signifikanten Unterschiede über die ...

4 CONCLUSION

Diese Arbeit befasste sich grundsätzlich mit der Frage, welche Anforderungen an ... gestellt werden. Der Lösungsvorschlag war... Benutzerstudien haben gezeigt, dass es einen signifikanten Unterschied... gibt. Diese Ergebnisse motivieren, um ...

4.1 Outlook

In dieser Arbeit haben wir uns mit ... beschäftigt. Durch zwei Benutzerstudien wurde festgestellt, dass sich die Aufteilung und Positionierung der Informationen innerhalb der Anzeige der beiden Gruppen aufgrund bla bla ändert. Das entwickelte Konzept ist zwar bla bla, müsste aber für eine Wiederholung der Studien bla bla zur Erhebung quantitativer Daten wie folgt angepasst werden:

- xxx
- yyy
- xzz

Des Weiteren wurde festgestellt, dass bla bla. Auch das müssten zukünftige Varianten besser berücksichtigen, z. B., indem sie bla bla. Für Studien sollte zudem eine neutrale(re) Umgebung gewählt werden. Somit sollte sichergestellt werden können, dass beispielsweise ein möglicher Bias der Marke des Fahrzeugs sich nicht auf das HMI-Konzept auswirkt...

4.2 Einschränkungen

Hinsichtlich Einschränkungen, die eine breite Anwendung der Ergebnisse verhindern, sind zwei getrennte Bereiche zu betrachten. Zum einen die Evaluation, welche sich nur auf den ersten Teil der Arbeit bezieht, zum anderen das ausgearbeitete Konzept für die Anzeige, das auf dem ersten Teil der Arbeit aufbaut.

Evaluation Bla bla.

Konzept Bla bla. Das wurde etwa in beschrieben...

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